

Mattia D'Urso

mattiadurso98@gmail.com | github.com/mattiadurso | scholar.google.com/mattiadurso | linkedin.com/mattiadurso
Citizenship: Italian (EU Citizen) | Languages: Italian (Native), English (C1 - Full Professional)

Education

Feb 2024 – Ph.D. in **Computer Science and AI**

Today at *Institute of Visual Computing, Graz University of Technology (Austria)*

Funded by Sony Europe and under the supervision of Prof. Friedrich Fraundorfer.

Topics: 3D Computer Vision · Machine Learning · Structure-from-Motion · Geometric Optimization

Oct 2020 – M.Sc. in **Computer Science and AI**

Dec 2022 at *Alpen-Adria-Universität of Klagenfurt (Austria)* and *University of Udine (Italy)*

- **Thesis:** Predictive maintenance of semiconductor testing equipment using Deep Learning approaches.

Oct 2017 – B.Sc. in **Computer Science**

Oct 2020 at *University of Udine (Italy)*

- **Thesis:** Classification of web users using shallow packet inspection and Deep Learning techniques.

Experiences

May 2026 – **ETH Zürich**, Zürich, *Visiting Researcher*

Oct 2026

- Supervised by Dr. Dániel Baráth and Prof. Marc Pollefeys at the CVG group.

Jun 2023 – **Canva**, Vienna, *Applied Scientist Intern*

Aug 2023

- Adapted Discrete Stable Diffusion paper for Layout Generation on a custom dataset.

- Improved by 20% the number of perfect samples generated by enhancing data and model.

Mar 2022 – **Infineon Technologies AG**, Villach, *Applied Scientist Intern*

Nov 2022

- Developed a predictive maintenance system based on time series using Deep Learning models.

Publications

Boosting 3D Foundation Models with Edge-based Pose Optimization

Mattia D'Urso, Christian Sormann, Mattia Rossi, Friedrich Fraundorfer

European Conference of Computer Vision (ECCV) 2026

TerraSky3D: Multi-View Reconstructions of European Landmarks in 4K

Mattia D'Urso, Yuxi Hu, Christian Sormann, Mattia Rossi, Friedrich Fraundorfer

3DMV @ CVPRW 2026 [Best Paper Award (Runner-Up)]

A Streamlined Attention-based Network for Descriptor Extraction

Mattia D'Urso, Emanuele Santellani, Christian Sormann, Mattia Rossi, Andreas Kuhn, Friedrich Fraundorfer

13th International Conference on 3D Vision (3DV) 2026

A LoD of Gaussians: Out-of-Core Training and Rendering for Seamless Ultra-Large Scene

Felix Windisch, Thomas Köhler, Lukas Radl, Mattia D'Urso, Michael Steiner, Dieter Schmalstieg,

Markus Steinberger. *SIGGRAPH 2026*

PyTorchGeoNodes: Enabling Differentiable Shape Programs for 3D Shape Reconstruction,

Sinisa Stekovic, Arslan Artykov, Stefan Ainetter, Mattia D'Urso, Friedrich Fraundorfer

Computer Vision and Pattern Recognition Conference (CVPR) 2025

Other Experiences

Sep 2025 – **Mentor at Lead The Future**

Today

- Selected as a mentor for top Italian STEM students (acceptance rate below 15%).